Finite-Interval Occupation Logarithm and Reference-Crossing Bounds

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Abstract

We establish a sharp scalar occupation-logarithm inequality, an explicit logarithmic envelope for the half-filled sine-kernel crossing nuclear norm, and finite-dimensional trace consequences for Wilson–Dirac interval covariances. An elementary change of variables and a two-region majorization yield an explicit crossing bound with leading coefficient one-half in its logarithmic integral estimate. Weighted Hilbert–Schmidt geometry gives a companion inequality for the reference modular discrepancy. These inequalities refine the constants in a directional positive-modular-trace enclosure and a transverse reference estimate. The Wilson positive reference-modular trace asymptotic requires the separate weighted argument of *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*.

1. Reference operators and hypotheses

Let $N = 2L \geq 2$ be even. Denote by B_N the half-filled sine-kernel compression, J_N its commuting Jacobi operator, and

$$W_N = B_N(I - B_N), \qquad K_N = \log(I - B_N) - \log B_N, \qquad E_N = K_N + N\eta(J_N/N).$$

The finite spectrum of B_N lies strictly in $(0, 1)$, and B_N and J_N commute. For the longitudinal and positive-majorant operators S_N, G_N, use the Wilson full-line relations

$$0 \leq 2G \leq I, \quad [G, P_0] = 0, \quad S = G(I - 2P_0)$$

and their interval compressions. The finite particle–hole staggering D satisfies $DB_ND = I - B_N$, $DG_ND = G_N$, $DS_ND = -S_N$. In this geometry the finite even-dimensional reference has no occupation eigenvalue $1/2$.

Write

$$M_N = \operatorname{Tr} G_N, \quad M_{2,N} = \operatorname{Tr}(W_N J_N^2), \quad T_{K,N} = \operatorname{Tr}(W_N K_N^2).$$

The Wilson-majorant and reference results used below are the finite mass identity

$$M_N = N\left[\frac{1}{2} - \frac{\arccos r}{\pi\sqrt{1 - r^2}}\right], \qquad r = \frac{\mu}{N + \mu}, \qquad M_N \leq \frac{\mu}{\pi},$$

and the reference estimates

$$\operatorname{Tr}\sqrt{W_N} \leq \frac{2}{\pi} I_N, \quad I_N = \int_0^\infty \frac{e^{-t}\sqrt{1 - e^{-2Nt}}}{1 - e^{-2t}}\, dt, \quad M_{2,N} \leq \frac{32}{\pi^2}(H_N + 1),$$

where $H_N = \sum_{j=1}^{N} j^{-1}$. These imported Wilson-specific inputs require their own separately established proofs; no step below substitutes an external classical reference for them.

2. Optimal occupation-logarithm constant

Theorem 1. There is a unique $x_* > 0$ with $x_* \tanh x_* = 2$. The best universal constant in

$$b(1-b)\log^2\frac{1-b}{b} \le C_{\log}\sqrt{b(1-b)}, \qquad 0 < b < 1,$$

is

$$\boxed{C_{\log} = \frac{2x_*^2}{\cosh x_*} = 2.12888831330963714924809207535747 6\ldots.}$$

Proof. With $x = \frac{1}{2}|\log((1-b)/b)|$, the ratio of the left side to $\sqrt{b(1-b)}$ is $2x^2/\cosh x$. Its derivative has sign $2-x\tanh x$. The latter expression changes sign exactly once because $x\tanh x$ is strictly increasing from zero to infinity. The ratio vanishes at both endpoints of the x-axis, and the unique maximum is attained by the two particle–hole conjugate occupations. □

Therefore, by the spectral theorem,

$$\boxed{T_{K,N} \le C_{\log}\operatorname{Tr}\sqrt{W_N}.} \tag{1}$$

3. Explicit reference-crossing integral

Theorem 2. For every integer $N \ge 2$,

$$\boxed{I_N \le F_N := \operatorname{arsinh}\sqrt{N-1} + \sqrt{N}\arcsin(N^{-1/2}).} \tag{2}$$

In particular,

$$F_N = \tfrac{1}{2}\log N + 1 + \log 2 + O(N^{-1}). \tag{3}$$

Proof. Set $u = e^{-2t}$ in the defining integral to obtain the exact identity

$$I_N = \frac{1}{2}\int_0^1 \frac{\sqrt{1-u^N}}{\sqrt{u}(1-u)}\,du.$$

Let $a = 1 - N^{-1}$. On $(0,a)$, $\sqrt{1-u^N} \le 1$. On $(a,1)$, Bernoulli's inequality $u^N \ge 1 - N(1-u)$ gives $\sqrt{1-u^N} \le \sqrt{N(1-u)}$. Consequently,

$$\begin{aligned} I_N &\le \frac{1}{2}\int_0^a \frac{du}{\sqrt{u}(1-u)} + \frac{\sqrt{N}}{2}\int_a^1 \frac{du}{\sqrt{u(1-u)}} \\ &= \operatorname{artanh}\sqrt{1-N^{-1}} + \sqrt{N}\arcsin(N^{-1/2}) \\ &= F_N. \end{aligned}$$

Expanding the two elementary functions at large N gives (3). □

For a completely explicit comparison with a logarithmic envelope, note that $\operatorname{arsinh}\sqrt{N-1} \le \frac{1}{2}\log N + \log 2$, and the function $\arcsin x/x$ is increasing for $0 < x < 1$. Consequently, for $N \ge 2$,

$$F_N \le \tfrac{1}{2}\log N + \log 2 + \frac{\pi}{2\sqrt{2}} < \frac{\log N + 2\sqrt{2} + e^{-1}}{1 - e^{-2}}.$$

The displayed inequalities are uniform in N and do not claim optimality of the integral majorant.

Corollary 3. Every finite interval satisfies

$$\operatorname{Tr}\sqrt{W_N} \le \frac{2F_N}{\pi}, \qquad T_{K,N} \le \frac{2C_{\log}F_N}{\pi}. \tag{4}$$

This does not assert asymptotic equality; the upper bound carries no inferred lower-bound coefficient.

4. Directional Wilson positive-trace enclosure

Let $P_N = \mathrm{Tr}(G_N|K_N|)$. The full-line projector relations give $0 \le G + S \le 2G$ and $G + S \le I - P_0$, and the inequalities persist under interval compression. For a reference eigenvector with occupation $b > 1/2$, denote $q = \langle v, G_N v\rangle$ and $s = \langle v, S_N v\rangle$. Then $u = q + s$ obeys $0 \le u \le 2q$ and $u \le 1 - b$. Particle–hole pairing yields

$$P_N - \mathrm{Tr}(S_N K_N) = 2 \sum_{b_j > 1/2} u_j |k_j| \ge 0.$$

Since $u_j \le \sqrt{2q_j(1-b_j)}$, $\sum_{b_j>1/2} q_j = M_N/2$, $b_j \ge 1/2$, and $T_{K,N} = 2\sum_{b_j>1/2} b_j(1-b_j)k_j^2$, Cauchy–Schwarz proves

$$\boxed{0 \le P_N - \mathrm{Tr}(S_N K_N) \le U_N := 2\sqrt{M_N T_{K,N}}.} \tag{5}$$

Corollary 4 (fully explicit error). With $C_{\log}$ from Theorem 1 and F_N from Theorem 2,

$$\boxed{U_N \le U_{N,\mu}^{\mathrm{elem}} := 2\sqrt{\frac{2C_{\log} M_N F_N}{\pi}}.} \tag{6}$$

For fixed positive μ, the bound has the asymptotic upper-envelope behavior

$$U_{N,\mu}^{\mathrm{elem}} = \frac{2}{\pi}\sqrt{C_{\log}\mu \log N}\,[1 + o(1)]. \tag{7}$$

The factor $O(\sqrt{\log N})$ is not improved to $O(1)$. Combining (5) with the independently proved cubic Jacobi envelope gives, with $C_\eta = 16\,\mathrm{arsinh}(1) - 4\pi$, $a_\mu = \sqrt{2/15}\,\mu^2/\pi$, $b_\mu = \mu/(3\pi)$,

$$V_{N,\mu} = C_\eta \left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right),$$

$$\boxed{P_N + \pi\,\mathrm{Tr}(S_N J_N) - U_{N,\mu}^{\mathrm{elem}} - V_{N,\mu} \le \mathrm{Tr}(S_N E_N) \le P_N + \pi\,\mathrm{Tr}(S_N J_N) + V_{N,\mu}.} \tag{8}$$

5. Weighted reference-modular variance

Let $A_\eta = 4\,\mathrm{arsinh}(1)$, so the established scaling-function bound is $|\eta(t)| \le A_\eta |t|$ for $|t| \le 1/2$. Because B_N, J_N, and their functions commute, the weighted Hilbert–Schmidt triangle inequality gives

$$\begin{aligned}\sqrt{\mathrm{Tr}(W_N E_N^2)} &= \|\sqrt{W_N}(K_N + N\eta(J_N/N))\|_2 \\ &\le \sqrt{T_{K,N}} + \sqrt{\mathrm{Tr}\{W_N[N\eta(J_N/N)]^2\}} \\ &\le \sqrt{C_{\log}\,\mathrm{Tr}\,\sqrt{W_N}} + A_\eta\sqrt{M_{2,N}}.\end{aligned}$$

Corollary 5. Consequently,

$$\boxed{\sqrt{\mathrm{Tr}(W_N E_N^2)} \le \sqrt{\frac{2C_{\log}F_N}{\pi}} + \frac{4\sqrt{2}A_\eta}{\pi}\sqrt{H_N + 1}.} \tag{9}$$

The established transverse pairing estimate $|\langle v_j, D_N v_j\rangle| \le 2\sqrt{q_j b_j(1-b_j)}$ now gives

$$\boxed{|\mathrm{Tr}(D_N E_N)| \le 2\sqrt{M_N}\left[\sqrt{\frac{2C_{\log}F_N}{\pi}} + \frac{4\sqrt{2}A_\eta}{\pi}\sqrt{H_N+1}\right].} \tag{10}$$

Both results are still $O_\mu(\sqrt{\log N})$. They rely on the Wilson pairing, crossing and second-moment theorems, not on a pointwise finite-size approximation of individual modular eigenvalues.

6. Scientific scope

The leading-order Wilson reduction and the exact Jacobi first moment imply the reformulation

$$\mathcal{D}_L(\mu) = P_N + \frac{\mu^2}{4} - \frac{\mu^2}{6}\log\frac{8}{\mu} + O_\mu(\sqrt{\log N}), \qquad N = 2L.$$

The finite-dimensional bounds here do not determine the limit of $P_N/\log N$. The separate leading entropy law of *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, combined with the positive-trace reduction above, establishes

$$\boxed{P_N = \frac{\mu^2}{6}\log N + o_\mu(\log N).}$$

The estimates established here concern occupation sums and reference crossing. The finite-part limit and uniform modular bound are established in *Finite Part of Wilson–Dirac Relative Entropy* and *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*, respectively. Their results do not assert a moving-edge interpolation expansion.

References

- NIST Digital Library of Mathematical Functions, §19.12, *Asymptotic Approximations*, https://dlmf.nist.gov/19.12 (classical elliptic-integral background).
- V. Eisler and I. Peschel, *Analytical results for the entanglement Hamiltonian of a free-fermion chain*, *Journal of Physics A* **50** (2017), 284003; DOI 10.1088/1751-8121/aa76b5 (commuting free-fermion entanglement-Hamiltonian background).
- Companion mathematical manuscripts for the Wilson-specific majorant, reference crossing, Jacobi moment and relative-entropy reduction, retained in the scientific collection.

Contributions and acknowledgments

Kirtus Justus directs the research program and publication decisions. ChatGPT (OpenAI) derived and checked the scalar and crossing estimates and their propagated consequences as AI-assisted mathematical work. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged as participants in the wider program, without attributing contributions to this manuscript unsupported by its provenance. No institutional endorsement, external peer review, or formal proof-assistant certification is implied.